

PIZZA SALES ANALYSIS USING SQL

Data Exploration & Business Insights from a
Pizza Sales Dataset

All
About
Pizza!



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ABOUT ME

- Aspiring Data Analyst | B.E. Electronics & Communication Engineering
- Background across full-stack development, embedded systems, and Generative AI training
- Currently building hands-on SQL & Power BI projects to strengthen data analytics skills
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Retrieve the total number of orders placed

```
select count(order_id) as total_orders from orders;
```

	total_orders
▶	21350



Calculate the total revenue generated from pizza sales

```
select  
round(sum(orders_details.quantity * pizzas.price),2) as total_sales  
from orders_details  
join pizzas on pizzas.pizza_id = orders_details.pizza_id
```

	total_sales
▶	817860.05



Identify highest priced pizza

```
select pizza_types.name, pizzas.price
from pizza_types
join pizzas on pizzas.pizza_type_id = pizza_types.pizza_type_id
order by pizzas.price desc limit 1;
```

name	price
The Greek Pizza	35.95



Identify the most common pizza size ordered

```
select pizzas.size, count(orders_details.order_details_id) as order_count
from pizzas
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizzas.size
order by order_count desc ;
```

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28



List the top5 most ordered pizza types along with their quantities

```
select pizza_types.name, sum(orders_details.quantity) as quantity
from pizza_types
join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity desc limit 5;
```

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371



Join the necessary tables to find the total quantity of each pizza category ordered

```
select pizza_types.category, sum(orders_details.quantity) as quantity
from pizza_types
join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category
order by quantity desc;
```

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050



Determine the distribution of orders by hour of the day

```
select hour(order_time) as hour, count(order_id) as order_count  
from orders  
group by hour(order_time);
```

hour	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399



Join relevant tables to find the category-wise distribution of pizzas

```
select category, count(name) from pizza_types  
group by category;
```

category	count(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9



Group the orders by date and calculate the average number of pizzas ordered per day

```
select round(avg(quantity)) as avg_pizza_ordered_per_day
from
(select orders.order_date, sum(orders_details.quantity) as quantity
from orders
join orders_details on orders.order_id = orders_details.order_id
group by orders.order_date) as order_quantity;
```

avg_pizza_ordered_per_day

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Determine the top 3 most ordered pizza types based on revenue

```
select pizza_types.name, sum(orders_details.quantity * pizzas.price) as revenue
from pizza_types
join pizzas on pizzas.pizza_type_id = pizza_types.pizza_type_id
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by revenue desc limit 3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5



Calculate the percentage contribution of each pizza type to total revenue

```
select pizza_types.category, (sum(orders_details.quantity * pizzas.price)/ (select  
round(sum(orders_details.quantity * pizzas.price),2) as total_sales  
from orders_details  
join pizzas on pizzas.pizza_id = orders_details.pizza_id) *100,2) as revenue  
from pizza_types  
join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id  
join orders_details on orders_details.pizza_id = pizzas.pizza_id  
group by pizza_types.category
```



Analyze the cumulative revenue generated over time

```
select order_date,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(orders_details.quantity * pizzas.price) as revenue  
from orders_details  
join pizzas on orders_details.pizza_id = pizzas.pizza_id  
join orders on orders.order_id = orders_details.order_id  
group by orders.order_date) as sales;
```

order_date	cum_revenue
2015-01-01	2713.85000000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05
2015-01-09	21525.4



Determine the top 3 most ordered pizza types based on revenue for each pizza category

```
select name, revenue
from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from(select pizza_types.category, pizza_types.name,
sum((orders_details.quantity * pizzas.price)) as revenue
from pizza_types
join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join orders_details on orders_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn<=3;
```

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75

